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Technical Approach

Functional Decomposition

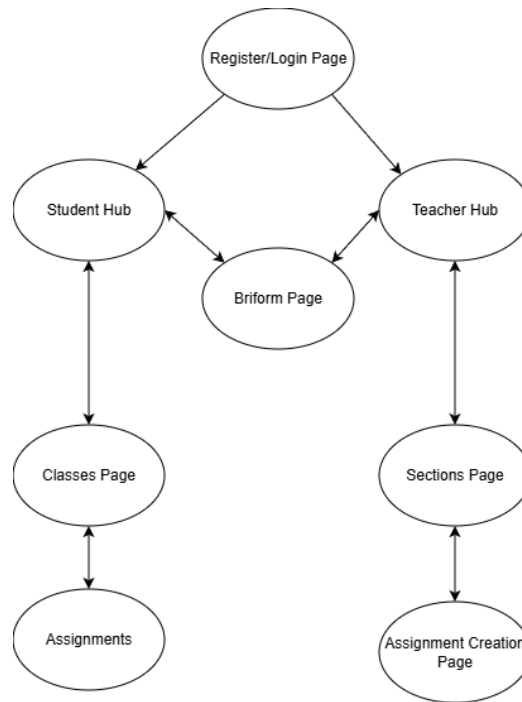


Figure 1: Technical Diagram

The first functional unit of the website is the register and login page. On this page, the user will be prompted to either log in to an existing account or to create a new account. If a user needs to create an account, they will be directed to do so while selecting whether they are a teacher or a student, and then will be redirected to the appropriate landing page. If a user already has an account and logs in, they will also be redirected to the appropriate landing page.

The student hub of our site is going to be where students end up after they log in. This will contain all of their classes as well as any assignments that are soon due. It will also have a redirect to the briform style page in case they want to make something independent from their classes. They will be able to view any saved projects they have completed, as well as be able

to manage them.

From the student hub, students will be able to access their individual classes through the classes page. This will allow them to go into their classes and see anything that has been posted by the instructor, as well as any lessons they need to complete.

Via the classes page, students will be able to move to an assignment page in order to be able to see all assignments for that class, as well as their due dates and any grades if the assignments have been graded.

When it comes to teachers, after logging in, they will be sent to the teacher hub. In this hub, they will be able to view all classes they are currently running, as well as being able to make new classes. They can also directly access the briform style page from here as well if they want to use the tool and create new projects, as well as save them. Teachers will also be able to access the sections page from the teacher hub.

The sections page is a per-class page that lets the teacher manage that class. From here, they can post announcements or other class-related posts. They will also be able to create sections that, when the students are on the classes page, they will be able to view the sections the teachers create. In these sections teacher can post material for students to learn from. The last thing the teacher can do on the sections page is to post and grade assignments for the students.

In the assignment creation page, the teacher will be able to make new assignments for their students to work on. These can be simple assignments, such as record and submit something, or more involved assignments where they have to use the Briform style tool to make and submit a project. When using the assignment creation page, the teacher can also set up how the assignment is going to be graded. They can decide whether the assignment will be graded automatically or if it will be manually graded by them after the students submit.

Finally, we have a Briform-style page that takes inspiration from the site Briformer. In this page, students and teachers will be able to make new projects in which they use a music audio file or a video to score the music and practice things like ear theory. They will also have the choice to save the project they finished, and if they want to continue a project, they can upload an existing saved project to resume where they left off.

Design Concepts, Evaluation, and Selection

Viktor Potter

One of the major design decisions for our product was whether assignments could be marked as autograded or manually graded by teachers, or if we would require teachers to grade their assignments with no option for autograding. For the choice of making teachers grade themselves with no option for autograding, it will be as simple as providing the teacher a box where they put the total possible score when creating an assignment, and then having them go back once a student submits an assignment to put in a grade themselves. When it comes to having an autograde system, we would have to make our own model for how assignments should be graded, as well as implement a way for teachers to adjust the autograding system. When it comes to these two alternatives for grading, the best decision for what we are making would be to only have manual grading enabled. As the time for our project is limited and along with all of the other functions that need to be built for the website, the time it would take to set up a proper autograding system would be too high compared to its overall importance to the project. The autograding feature may also be something that could cause teachers' headaches if we do not implement it properly, or if there are bugs within the autograding system that we fail to find. The time that would be spent making this feature would be utilized much better for optimizing the main components of the site and enhancing the user experience through a well-polished site instead of an autograding system.

Deliverables

Our project deliverables can be divided into three main phases: Development, Testing/Feedback, and Deployment. The specific deliverables for each category are outlined below:

Development

- Implement account structure using Supabase and integrate these accounts with YouTube/Spotify accounts

- Complete "Neo-Briform" tool that allows students to diagram form and phrase structure in a more intuitive and user-friendly way than Briform
- Complete song-to-score functionality that allows students to test their ear training skills against music that they are familiar with

Testing/Feedback

- Select a preliminary test group of students to use the tool and provide feedback
- Create a Google Form that test users can fill out to provide feedback on the tool
- Conduct at least one round of testing and feedback collection from the preliminary test group
- Use given feedback to improve the tool and fix any bugs or issues that arise

Deployment

Note: These tasks will mostly fall on Connor as he has important connections in the College of Music

- Coordinate meetings with Theory Department head and other relevant faculty to present the tool and discuss its use cases in our curriculum
- If necessary, make any final adjustments to the tool based on faculty feedback
- Potentially coordinate meeting with Dean of the College of Music to discuss launching the tool

Project Management

The timeline below includes major milestones and expected completion time frames. It is important to note that this timeline does not give specific expectations but simply serves as a baseline expectation for how we will progress.

Proposed Project Timeline

Note: The bold lines in the table serve to separate our milestones into three main phases: Development, Testing/Feedback, and Deployment

Table 1: Proposed Project Timeline

Milestone	Target Date
Basic implementation of Briform clone and score reduction functionality	End of January
Continued work on refining tool and preparing for testing	Early/Mid February
Preliminary test group is selected and asked to complete an assignment that utilizes the tool	End of February
Begin improving tool based on feedback received from test group	Spring Break
Continue improving based on feedback and desired features and begin communicating with College of Music faculty	Early April
Fix remaining bugs, polish features, and continue communications with College of Music faculty	Mid/Late April
Production Ready	Early May

Budget

Because we have no financial costs associated with our project, our budget will outline how much time we expect to allocate towards specific tasks. Our goal is to work 30-40 hours per week until completion. In the table below, our estimated workload is shown as a percentage of our weekly expectation, and three percentages are listed for each of the three phases of our project.

Estimated Budget

Table 2: Estimated Budget

Tasks	Expected Workload (%)
Development Deliverables	1: 90%, 2: 40%, 3: 10%
Feedback Deliverables	1: 10%, 2: 60%, 3: 10%
Deployment Deliverables	1: 00%, 2: 00%, 3: 80%